

Binance Futures Trading Bot – Report

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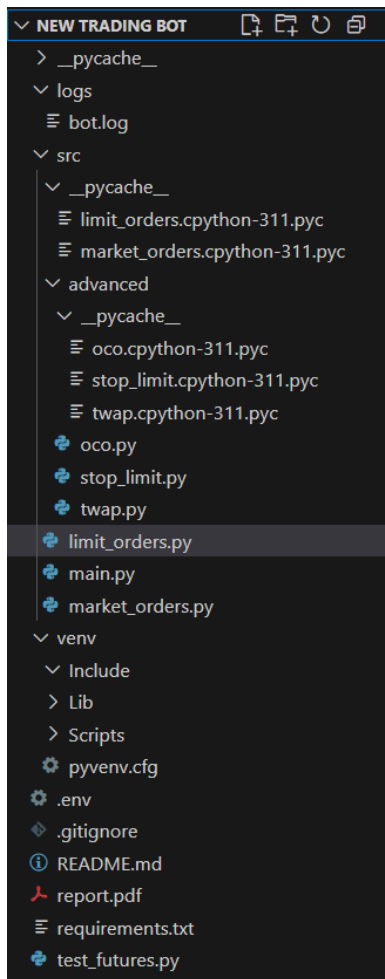
1. Project Overview

The Binance Futures Trading Bot is a **CLI-based Python application** designed for **USDT-M Futures Testnet**. It allows users to place **market and limit orders**, as well as **advanced order types** like Stop-Limit, OCO, and TWAP.

Objectives of the project:

- Automate trading on Binance Futures Testnet safely.
 - Implement both **basic and advanced order types**.
 - Validate user inputs for trading pairs, order sides, quantities, and prices.
 - Maintain structured **logs of all order executions and errors**.
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2. Project Structure



3. Setup Instructions

1. Clone the repository or extract the .zip file:

```
git clone [your_repo_url]
```

```
cd new_binance_bot
```

2. Create a Python virtual environment:

```
python -m venv venv
```

```
venv\Scripts\activate # Windows
```

```
source venv/bin/activate # macOS/Linux
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Create a .env file in the project root with your **Binance Testnet API keys**:

```
BINANCE_API_KEY=your_testnet_api_key
```

```
BINANCE_API_SECRET=your_testnet_api_secret
```

4. CLI Usage and Examples

4.1 Market Order

- **Input Example:**

Enter trading pair: BTCUSDT

Enter side: BUY

Enter order type: MARKET

Enter quantity: 0.001

- **Expected Output:** Market order is executed immediately at the current market price.
- **Screenshot:** *Add your screenshot here*

4.2 Limit Order

- **Input Example:**

Enter trading pair: BTCUSDT

Enter side: SELL

Enter order type: LIMIT

Enter quantity: 0.001

Enter limit price: 60000

- **Expected Output:** Limit order is placed at the specified price. Execution occurs when market reaches that price.
- **Screenshot:** *Add your screenshot here*

4.3 Stop-Limit Order (Advanced)

- **Input Example:**

Enter trading pair: BTCUSDT

Enter side: BUY

Enter order type: STOP-LIMIT

Enter quantity: 0.001

Enter stop price: 55000

Enter limit price: 55100

- **Explanation:** Triggers a limit order once the stop price is reached. Useful for automating entry strategies.

- **Screenshot:** *Add your screenshot here*
-

4.4 OCO Order (Advanced)

- **Input Example:**

Enter trading pair: BTCUSDT

Enter side: SELL

Enter order type: OCO

Enter quantity: 0.001

Enter take-profit price: 62000

Enter stop-loss price: 58000

- **Explanation:** One-Cancels-the-Other places a take-profit and stop-loss simultaneously; execution of one cancels the other.
 - **Screenshot:** *Add your screenshot here*
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4.5 TWAP Order (Advanced)

- **Input Example:**

Enter trading pair: BTCUSDT

Enter side: BUY

Enter order type: TWAP

Enter quantity: 0.01

Enter number of slices: 5

Enter delay between slices in seconds: 10

- **Explanation:** Splits a large order into multiple smaller orders to minimize market impact over time.

- Screenshot:

```
(venv) PS C:\Users\Admin\OneDrive\Desktop\new trading bot> python src/main.py
✓ Connected to Binance Futures Testnet!

--- New Order ---
Enter trading pair (e.g., BTCUSDT): BTCUSDT
Enter side (BUY/SELL): BUY
Enter order type (MARKET, LIMIT, STOP-LIMIT, TWAP, OCO): TWAP
Enter quantity: 0.01
Enter number of slices (e.g., 5): 5
Enter delay between slices in seconds (e.g., 10): 10
⚡ TWAP order: 0.01 BTCUSDT in 5 slices (0.002 per slice)
✓ Slice 1/5 executed: 0.002 BTCUSDT
✓ Slice 2/5 executed: 0.002 BTCUSDT
✓ Slice 3/5 executed: 0.002 BTCUSDT
✓ Slice 4/5 executed: 0.002 BTCUSDT
✓ Slice 5/5 executed: 0.002 BTCUSDT
✓ TWAP order complete!
✓ Order processed. Check logs for details.
Place another order? (y/n): n
(venv) PS C:\Users\Admin\OneDrive\Desktop\new trading bot> █
```

5. Logging

- All bot activity is logged to logs/bot.log.
- Example log entries:

```
logs > botlog
1 2025-10-23 19:02:44,850 - INFO - Bot initialized with Binance Futures Testnet
2 2025-10-23 19:08:46,508 - INFO - Bot initialized with Binance Futures Testnet
3 2025-10-23 19:15:01,860 - INFO - Bot initialized with Binance Futures Testnet
4 2025-10-23 19:17:18,482:INFO:Market order successful: {'orderId': 6751946609, 'symbol': 'BTCUSDT', 'status': 'NEW', 'clientOrderId': 'x-Cb7ytek1962166fc415515c2c88fb', 'price': '0.00', 'avgPrice': '0.00', '
5 2025-10-23 19:25:03,839:INFO:Bot initialized with Binance Futures Testnet
6 2025-10-23 19:26:12,198:ERROR:Error placing Stop-Limit order: APIError(code=-4164): Order's notional must be no smaller than 100 (unless you choose reduce only).
7 2025-10-23 19:27:17,882:ERROR:Error placing limit order: APIError(code=-4164): Order's notional must be no smaller than 100 (unless you choose reduce only).
8 2025-10-23 19:29:22,315:INFO:Bot initialized with Binance Futures Testnet
9 2025-10-23 19:30:22,139:ERROR:Error placing limit order: APIError(code=-4005): Quantity greater than max quantity.
10 2025-10-23 19:32:25,498:INFO:Bot initialized with Binance Futures Testnet
11 2025-10-23 19:32:46,481:ERROR:Error placing limit order: APIError(code=-2019): Margin is insufficient.
12 2025-10-23 19:33:03,499:ERROR:Error placing limit order: APIError(code=-2027): Exceeded the maximum allowable position at current leverage.
13 2025-10-23 19:33:19,636:ERROR:Error placing limit order: APIError(code=-4005): Quantity greater than max quantity.
14 2025-10-23 19:34:01,272:INFO:Limit order successful: {'orderId': 6753853079, 'symbol': 'BTCUSDT', 'status': 'NEW', 'clientOrderId': 'x-Cb7ytek3a24fc6bddd4f28c4a70e14', 'price': '50000.00', 'avgPrice': '0.00', '
15 2025-10-23 19:46:58,250 - INFO - TWAP slice 1 executed: {'orderId': 6755290253, 'symbol': 'BTCUSDT', 'status': 'NEW', 'clientOrderId': 'x-Cb7ytek35d191f4075872643d37711', 'price': '0.00', 'avgPrice': '0.00', '
16 2025-10-23 19:47:08,718 - INFO - TWAP slice 2 executed: {'orderId': 6755310232, 'symbol': 'BTCUSDT', 'status': 'NEW', 'clientOrderId': 'x-Cb7ytek3a21dcc5712a10d2400cd49', 'price': '0.00', 'avgPrice': '0.00', '
17 2025-10-23 19:47:18,896 - INFO - TWAP slice 3 executed: {'orderId': 6755329661, 'symbol': 'BTCUSDT', 'status': 'NEW', 'clientOrderId': 'x-Cb7ytek3ae988be8cf229930a0aa3', 'price': '0.00', 'avgPrice': '0.00', '
18 2025-10-23 19:47:29,096 - INFO - TWAP slice 4 executed: {'orderId': 6755347818, 'symbol': 'BTCUSDT', 'status': 'NEW', 'clientOrderId': 'x-Cb7ytek35ccf3d65fd20172e976055', 'price': '0.00', 'avgPrice': '0.00', '
19 2025-10-23 19:47:39,270 - INFO - TWAP slice 5 executed: {'orderId': 6755364936, 'symbol': 'BTCUSDT', 'status': 'NEW', 'clientOrderId': 'x-Cb7ytek38eb7ebf2109dab0cfcf320', 'price': '0.00', 'avgPrice': '0.00', '
20
```

- Logs include timestamps, order details, and any errors for troubleshooting.

6. Input Validation and Error Handling

- Validates:
 - Trading pair format (e.g., BTCUSDT)
 - Order side (BUY or SELL)
 - Quantity (must meet Binance minimum notional)

- Price thresholds for limit and stop orders
- Handles API exceptions gracefully and logs errors:

```
--- New Order ---
Enter trading pair (e.g., BTCUSDT): BTCUSDT
Enter side (BUY or SELL): BUY
Enter order type (MARKET, LIMIT, STOP-LIMIT): LIMIT
Enter quantity: 200
Enter limit price: 50000
✗ Error: APIError(code=-2019): Margin is insufficient.
Please enter a larger quantity or adjust price to meet minimum notional (100 USDT).
(100 USDT).
Enter quantity: 10000
Enter limit price: 50000
✗ Error: APIError(code=-4005): Quantity greater than max quantity.
Please enter a larger quantity or adjust price to meet minimum notional (100 USDT).
Enter quantity: 0.002
Enter limit price: 50000
✅ Limit order placed successfully!
{'orderId': 6753853079, 'symbol': 'BTCUSDT', 'status': 'NEW', 'clientOrderId': 'x-Cl
: '0.00000', 'timeInForce': 'GTC', 'type': 'LIMIT', 'reduceOnly': False, 'closePosi
IT', 'priceMatch': 'NONE', 'selfTradePreventionMode': 'EXPIRE_MAKER', 'goodTillDate
Place another order? (Y/N): N
Exiting bot. Goodbye!
```

7. References

- [Binance Futures API Documentation](#)
- Python-Binance Library
- Historical Data (Optional): [Download Link](#)
- Fear & Greed Index (Optional Bonus): [Download Link](#)

8. Conclusion

This project demonstrates:

- Safe and automated trading on Binance Futures Testnet.
- Support for **basic and advanced order types**.
- Proper **logging, validation, and error handling**.
- A simple, easy-to-use **CLI interface** for experimentation.

Note: Always use Testnet to avoid real financial risk.